

List the 6 steps to establish experience assumptions.

“I DARDM” to establish experience assumptions!

1. **I**dentify assumptions needed
2. **D**etermine structure of each assumption
3. **A**nalyze experience and trends
4. **R**eview assumptions for reasonableness, consistency
5. **D**ocument assumptions
6. **M**onitor experience and update assumptions

List primary types of assumptions for experience studies.

1. Obligation (liability) assumptions

- Mortality, lapse, expense, etc.

2. Asset assumptions

- Investment income rate, capital gains rate, defaults, etc.

3. Scenario assumptions

- Deterministic vs. stochastic interest rates
- Sensitivity testing

How to determine assumption structure?

Determine experience classes

- Groups of policies with same assumption
- Similar type, structure, marketing objectives

Key principles when deciding complexity

- Reflect differences in actual experience
- Use objective definitions
- Be practical and cost effective

List considerations when analyzing experience and trends.

Source: ILA101-106: Experience Assumptions

1. **Evaluate credibility**

- Quantity of data, homogeneity, reasonableness

2. **Evaluate data quality**

- Alternative sources?
- Appropriate? Comprehensive enough?

3. **Actual vs. similar experience**

- Use actual if available and credible

4. **Reflect trends**

- Example: mortality improvement

5. **Reflect company and external factors**

- UW, investment policy, other business practices

6. **Sensitivity test assumptions**

List checks for reviewing assumptions for reasonableness and consistency.

Consistency Checks

- Inflation consistent with investment earnings?
- Mortality anti-selection and lapses

Validation Checks

- Static (starting reserves)
- Dynamic (projected reserves)

How to document assumptions?

Source: ILA101-106: Experience Assumptions

- **Actual assumptions**
 - Value, applicable class
- **Data used**
 - Source, values, concerns, adjustments, reliance on others
- **Methods used to develop assumptions**
 - E.g. credibility method
- **How to use assumption**
 - Pricing vs. CFT
 - Sensitivity testing
 - Regulatory requirements

List key steps to determine mortality assumption.

1. **Determine Structure**

- ▶ Example: S&U ALB table varying by M/F, SM/NS

2. **Analyze experience**

- ▶ Do mortality study (e.g. 5-year CY study)
- ▶ Develop expected mortality rates from study
- ▶ Assess credibility of expected assumption
- ▶ Adjust mortality rates
 - ▶ Trends (improvement, etc.)
 - ▶ Anti-selection – conservation of deaths
 - ▶ Blend with similar/industry experience if credibility low
 - ▶ Adjust for differences in UW, distribution, market, etc.
 - ▶ Blend M/F rates into aggregate rates (if needed)

3. **Monitor experience** (A/E ratios)

More detail on these steps in other flash cards

What is the structure of a mortality assumption?

Source: ILA101-106: Experience Assumptions

- Select and ultimate common
- Possible variations
 - ▶ Risk class
 - ▶ Selection process (type of UW)
 - ▶ Size of policy (bigger face → lower mortality)
 - ▶ Market method (direct, agent, etc.)
- Mortality improvement
- Age last birthday (ALB) vs. age nearest birthday (ANB)

$$q^{\text{ALB}}(x) = 0.5 \times [q^{\text{ANB}}(x) + q^{\text{ANB}}(x + 1)]$$

$$q^{\text{ANB}}(x) = 0.5 \times [q^{\text{ALB}}(x) + q^{\text{ALB}}(x - 1)]$$

How to analyze mortality experience using confidence intervals?

Source: ILA101-106: Experience Assumptions

Mean and variance for expected claims E :

$$E = nq$$

$$\text{Var}(E) = npq$$

Confidence interval for expected claims and the expected mortality rate q

$$95\% \text{ C.I. For } E : E \pm 1.96\sqrt{\text{Var}(E)}$$

$$95\% \text{ C.I. For } q : \frac{95\% \text{ C.I. For } E}{n}$$

What are the 2 main types of mortality studies, and how are different risk classes handled?

Source: ILA101-106: Experience Assumptions

Typically done on standard issues, with separate studies for:

1. Non-routine UW
2. Conversions → usually higher mortality
3. Sub-standards
4. Non-forfeiture provisions (ETI, RPU)
5. Multiple-life policies

Two types of studies:

1. Calendar year
 - ▶ Look at activity for a single calendar year
 - ▶ Must account for new policies, withdrawals, and deaths
2. Anniversary-to-anniversary
 - ▶ Simpler because study year coincides with policy year
 - ▶ Done on an issue year / duration basis

How to determine calendar year mortality exposure?

$$A + N = W + D + B$$

Situation	Exposure	
(A, W)	s	Existing policy withdraws at age $x + s$
(A, D)	1	Existing policy dies in calendar year
(A, B)	1	Existing policy survives entire calendar year
(N, W)	$s - r$	New policy enters at age $x + r$, then withdraws at age $x + s$
(N, D)	$1 - r$	New policy enters at age $x + r$, then dies in calendar year
(N, B)	$1 - r$	New policy enters at age $x + r$ and survives entire calendar year

How to determine mortality rate from a mortality study?

Balducci assumption:

$$q(x) = \frac{\text{Deaths}}{\text{Total Exposure}} = \frac{D}{A + (1 - r)N - (1 - s)W}$$

Can be done at policy level or in aggregate

- Policy level data
 - Sum up individual exposures
- Aggregate data (i.e. you just have the totals)
 - Assume new issues and withdrawals occur mid-year ($s = r = 0.5$)

$$q(x) = \frac{\sum D}{\sum A + 0.5 \sum N - 0.5 \sum W}$$

How to enhance credibility of mortality studies and develop confidence intervals for A/E ratios?

Credibility Considerations

1. Multi-year studies have higher credibility
 - ▶ Problem: Too many years may hide trends (5 years is good)
2. Credibility usually lower for amount-based studies
3. A/E ratios useful for assessing credibility and tracking trends

Actual-to-Expected (A/E) Ratios

	By Count	By Amount
E	$\sum q(i)$	$\sum A(i)q(i)$
Var	$\sum q(i)[1 - q(i)]$	$\sum [A(i)]^2 q(i)[1 - q(i)]$

$$95\% \text{ C.I. for Claims (CIC): } E \pm 1.96\sqrt{\text{Var}(E)}$$

$$95\% \text{ C.I. for A/E Ratio: } \frac{\text{CIC}}{E}$$

What is the conservation of deaths principle?

$$q_{[x]+t} = w^{AS} q_{[x+r]+t-r} + (1 - w^{AS}) q_{[x]+t}^{AS}$$

Total mortality = weighted average of:

1. Mortality of the “select” lives that lapse in duration r
2. Mortality of the lives that **stay with the original policy**

Key pricing concept: If you don't account for selective lapsation, you will under-price the product

List at least 4 variations in lapse assumption structure.

Source: ILA101-106: Experience Assumptions

1. Product design

- ▶ Term: High (shock) lapses when ART period begins
- ▶ Annuities: High (shock) lapses after SC period
- ▶ Permanent insurance: High 1st-year lapses

2. Distribution channel

- ▶ Brokerage: Higher lapses
- ▶ Agency: Higher lapses if agent quality is poor

3. Policy size

- ▶ Small policies: high early lapses
- ▶ Large policies: high later lapses

4. Premium mode

- ▶ Lapses generally occur on premium due dates
- ▶ Flexible premium: assume uniform monthly

5. Product type

- ▶ Deferred annuities: more sensitive to lapse rates than life insurance

6. Conservation program effectiveness

How to determine exposure for lapse study and credibility of lapse assumption?

Lapse Experience Credibility (Confidence Interval)

$$E \pm 1.96\sqrt{\text{Var}(E)}$$

Lapse Studies

$$w(x) = \frac{W}{A + (1 - r)N - (1 - s)D}$$

- Mechanically the same as a mortality study – death and lapse trade places
- Lapses get a full year of exposure

List specific types of lapses.

Source: ILA101-106: Experience Assumptions

1. Termination on:
 - ▶ Failure to pay premium (term)
 - ▶ Full cash surrender
 - ▶ Partial cash surrender
 - ▶ Policy loan $>$ CSV
2. Nonforfeiture transfers – ETI, RPU
3. Term conversions
4. Premium persistency – actual premium as % of target premium (UL)

Describe interest rate assumption structure: Deterministic vs. Stochastic and Policy Loans.

Investment Assets

- **Deterministic**
 1. Portfolio average
 2. Investment generation
- **Stochastic**
 - Useful for more important risks
 - Rates can vary with time, asset class, quality, and credit risk

Policy Loans

- Can be modeled as assets or negative liabilities
- Net of policy loan expenses
- Utilization rate

Write formulas for interest rates on book value vs. market value basis.

Book Value Basis:

$$I = Ai + (B - A - I) \left(\frac{i}{2} \right) \Rightarrow i = \frac{2I}{A + B - I}$$

Market Value Basis:

$$r = \frac{B - A - C}{A + \frac{C}{2}}$$

Write formulas for time-weighted vs. dollar-weighted rates.

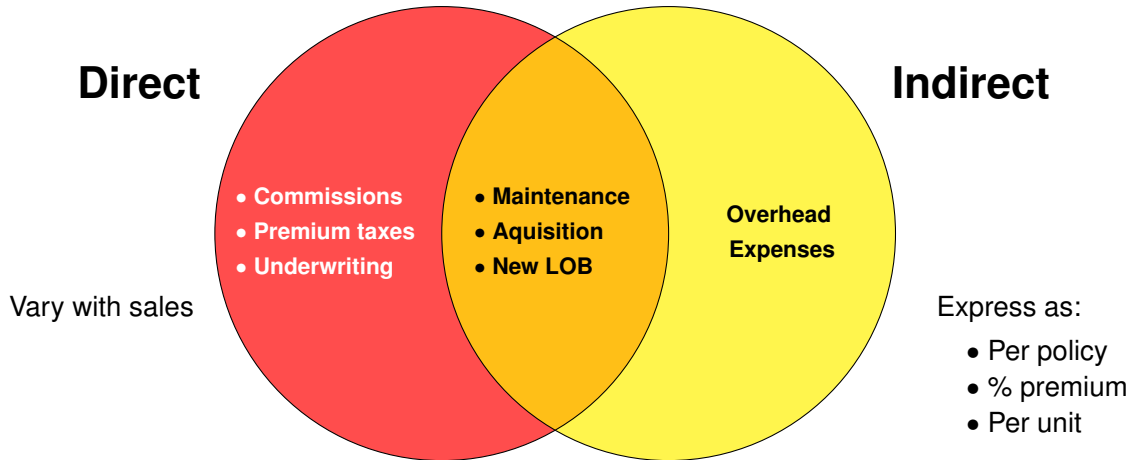
Time-weighted return

$$r = [(1 + r_1)(1 + r_2) \cdots (1 + r_n)]^{(1/n)} - 1$$

Dollar-weighted return

$$\text{Value}_t = CF_{t-1}(1 + r) + CF_{t-2}(1 + r)^2 + \cdots + CF_{t-n}(1 + r)^n$$

Distinguish between direct and indirect expenses.



How to determine exposure in an expense study?

Key Concepts

1. **Goal:** Develop policy count base for per policy expenses
2. **How:** Count the number of policy years that **start in the study's calendar year**
 - ▶ BOYs "crossed" in CY = $A + N - W/2$
 - ▶ Mid-years "crossed" in CY = $A + N/2 - W/2$

$$\text{Exposure Count} = \begin{cases} \frac{A + B + N}{2} & \text{for BOY expenses} \\ \frac{A + B}{2} & \text{for mid-year expenses} \end{cases}$$

Assumes withdrawals and new issues occur mid-year

List methods for allocating expenses.

Source: ILA101-106: Experience Assumptions

1. Transaction count (e.g. number of premium payments)
2. Transfer costs (employee benefit cost per employee)
3. Employee time spent on activities
4. Index-based allocation (policy count or premium)

Define:

- Study period
- Study population
- Cell
- Experience rates
- Rate types: Decrement rates, Utilization rates

Source: Experience Study Calculations

- **Study period** – period of time covered
- **Study population** – insureds, employees, etc.
- **Cell** – combinations of data dimensioned by issue age, sex, smoker/nonsmoker, policy year, etc.
- **Experience rates** – mortality rates, lapse rates, etc. for each cell
- **Rate type**
 - **Decrement rates** = probabilities from 0 to 1
 - Mortality, morbidity, lapse, etc.
 - **Utilization rates** $\neq$ probabilities (can exceed 1)
 - Withdrawal rates, option election rates, etc.

Calculate E_X reflecting **withdrawal rates** using Balducci Hypothesis:

- For individual calculations
- For grouped calculations

Source: Experience Study Calculations

$$E_x = \begin{cases} \ell_x - \sum_{i=1}^{w_x} (1 - t_i) = (\ell_x - w_x) + \sum_{i=1}^{w_x} t_i & \text{for individual calculations} \\ \ell_x - \frac{1}{2} w_x & \text{for grouped calculations} \end{cases}$$

Formula for exposure-weighted average mortality rate over an N -year period starting at age x

Source: Experience Study Calculations

$$\begin{aligned} q &= \frac{\sum_{t=0}^N E_{x+t} q_{x+t}}{\sum_{t=0}^N E_{x+t}} \\ &= \frac{\sum_{t=0}^N d_{x+t}}{\sum_{t=0}^N E_{x+t}} \end{aligned}$$

Amount-weighted studies: **individual amount weights**

Individual amount weights

- Multiply each life year exposure by policy's DB
- Sum amount exposures and calculate q_x 's in the usual way

Amount-weighted studies: **grouped amount weights**

Grouped amount weights

- The ℓ_x terms $\rightarrow \ell_x \times \text{Average DB In Force}$
- The d_x terms $\rightarrow d_x \times \text{Average DB Paid on Death}$
- The w_x terms $\rightarrow w_x \times \text{Average DB on Withdrawn Policies}$
- Sum amount exposures and calculate q_x 's in the usual way

Relationship between amount-weighted q and life-weighted q ?

Source: Experience Study Calculations

Amounted-weighted q will be $>$ life-weighted q if average DB paid on death $>$ average DB in force

A/E ratio at **individual** level?

Source: Experience Study Calculations

The A/E ratio at each age x is:

$$(A/E)_x = \frac{d_x}{d_x^e} = \frac{q_x E_x}{q_x^e E_x} = \frac{q_x}{q_x^e}$$

where e denotes expected values, and the others are actuals

A/E ratio at **aggregate** level?

Source: Experience Study Calculations

The A/E ratio at an aggregate level is:

$$q^e = \frac{\sum E_x q_x^e}{\sum E_x} = \frac{\sum d_x^e}{\sum E_x}$$
$$(A/E) = \frac{d}{d^e} = \frac{\sum d_x}{\sum d_x^e} = \frac{q}{q^e}$$

List uses of A/E analysis.

Source: Experience Study Calculations

- Compare actual mortality or lapse experience to expected amounts
- Develop best estimate assumptions as multiple of expected amounts

$$\text{Best Estimate Rate for Age } x = (A/E) \times q_x^e$$

- Can be used for valuation, risk management, financial planning, etc.

Formulas for **frequency** and **severity**?

Source: Experience Study Calculations

$$f_x = \frac{n_x}{E_x} = \text{average claim frequency}$$

$$s_x = \frac{C_x}{n_x} = \text{average claim amount}$$

- n_x = number of claims incurred at age x
- C_x = total claim amount incurred at age x
- E_x = central exposure (or initial)

Loss ratio formula?

Source: Experience Study Calculations

Loss ratio = cost of claims as % of premiums paid (P_x)

$$LR_x = \frac{C_x}{P_x}$$

Assume: M_x = max w/d allowed — W_x = total actual w/d amount — n_x = # of contracts who took w/d

Define formulas for:

- Count-based exposure (excludes deaths and lapses)
- Withdrawal frequency
- Average size (severity) of w/d taken
- Average w/d taken as % of max w/d

Source: Experience Study Calculations

$E_x = \ell_x - d_x - w_x$ = count-based exposure (excludes deaths and lapses)

$f_x = \frac{n_x}{E_x}$ = withdrawal frequency

$s_x = \frac{W_x}{n_x}$ = average size (severity) of withdrawals taken

$u_x = \frac{s_x}{M_x}$ = average withdrawal taken as % of the max withdrawal

Note: little w is still a lapse rate here; big W is what we're studying!

List distortions caused by amount-weighted calculations.

Source: Experience Study Calculations

1. **One or few large amounts can distort numerator**

- ▶ Not easy to fix after the fact (cap max amount allowed in study)

2. **Risks or behaviors can differ significantly by amount**

- ▶ Small policies may have less commitment
- ▶ Large policies may be used to defraud
- ▶ Solution: add size band to cells

List advantages of multi-year studies.

Source: Experience Study Calculations

- Higher credibility
- Less distortion caused by reporting lags
- Can populate many more cells
- Allows study of trends if study-year or CY variable is included

List 4 examples of reporting lags.

Source: Experience Study Calculations

- Death claims often not reported for months after death
- LTCI claims not reported until after elimination period
- “Shoe boxing” – lag between direct insurers and reinsurers
- Life annuities sometimes continue paying after death

List 2 solutions for reporting lags.

Source: Experience Study Calculations

- Wait a few months after study period before gathering data
- Develop IBNR claims estimates from previous studies

List 3 examples of non-uniform events.

Source: Experience Study Calculations

- Lapses often vary by month (e.g. PLT period)
- Prorated mortality rates tend to underestimate deaths early in the year at old ages
- LTCI claims immediately following claim

List 3 ways to compensate for non-uniform distributions.

Source: Experience Study Calculations

1. Use shorter interval of time
2. Apply adjustment factors
3. Use constant force mortality assumption for old ages

List considerations for inclusion of partial policy years in decrement studies.

Source: Experience Study Calculations

- **OK when event is evenly distributed over the year**
 - ▶ Common for biologically-driven rates (mortality, morbidity, etc.)
 - ▶ Allocating PYs between CYs is also common
- **Otherwise, partial policy years will distort results**
 - ▶ Behavioral rates usually not evenly distributed
 - ▶ Lapse, withdrawal, option election, etc.
 - ▶ Do not allocate PYs between CYs

What is the **annual** exposure method?

Annual exposure method – consistent with Balducci Hypothesis

- Exposes decrements to end of next anniversary (even if beyond study period)
- Disadvantage: overstates rates in first partial year; understates rates in final partial
- Why exposure is extended beyond study period:
 - ▶ Rate is based on probability
 - ▶ Captures exposure missed by initial partial year
 - ▶ Also true of distributed exposure method

What is the **distributed** exposure method?

Distributed exposure method – assumes decrement is uniformly distributed

- Allocates exposure and decrements by both PY and CY
- More accurate when underlying rates are increasing; otherwise less accurate
- Produces same total PY exposure as annual method, but CY allocation differs

Algorithm for calculating distributed exposure?

Source: Experience Study Calculations

1. Calculate annual exposure in usual way
2. First-half exposure = PY t exposure in CY:
 - Expose deaths and surviving policies from policy anniversary to end of CY
 - Expose other decrements (lapse, etc.) from policy anniversary to decrement date
3. Second-half exposure = Total Annual Exposure – First-Half Exposure
 - This will be allocated to next CY

In plain English: switch the partial year formulas in the annual exposure method

- *Move final partial year deaths to next study period's first partial year*

List product-related considerations that may affect experience study calculations.

Source: Experience Study Calculations

- **Deaths in premium grace period** (30–60 days)
 - Actual DB paid is net of overdue premiums, but use full DB in mortality study
 - Expose lapses to beginning of grace period
- **Compromised and denied claims** – don't count as death claims in mortality study
- **Study variables changing over time** (admin system changes)
 - Admin systems may change plan codes or billing frequencies (distorts exposure)
- **Reinsured amounts** – can study amounts net of reinsurance if material enough
- **Substandard and uninsurable lives** – typically excluded

List at least 4 claim characteristics shared by LTCI and DI products.

Source: Experience Study Calculations

1. Involve multiple payments that can last for months or years
2. Usually paid monthly
3. Subject to elimination period before claim payments start (up to 1 year)
4. Usually limited in some way (e.g. max age for DI, max total benefits, etc.)
5. Paid when insured meets certain conditions
6. Claims stop when insured recovers but can start again

List types of morbidity studies for LTCI and DI:

- Claim incidence studies
- Claim severity studies
- Claim termination studies
- Benefit utilization rate studies

Source: Experience Study Calculations

1. **Claim incidence studies**

- ▶ Claim incidence rates = annual probabilities
- ▶ Usually done on “all claims”
- ▶ Usually based on annual exposure method

2. **Claim severity studies**

$$\text{Average length of claim} = \frac{\text{Total Months of Claims}}{\text{Number of Claims}}$$

$$\text{Average cost per claim} = \frac{\text{Total Claims Paid}}{\text{Number of Claims}}$$

3. **Claim termination studies** – perform separate studies for recovery and death

- ▶ DI claims terminate when proof of disability not submitted
- ▶ LTC claims terminate when requests for reimbursement stops
- ▶ Termination rates are highest and most volatile for new claims

4. **Benefit utilization rate studies** (LTCL only)

$$\text{Benefit Utilization Rate} = \frac{\text{Actual Benefits}}{\text{Max Benefits}}$$

List 4 **DI-specific** considerations for experience studies.

Source: Experience Study Calculations

- Elimination periods – claims usually reported after elimination period
- Partial disability benefits – reduces average cost per claim
- Recovery followed by relapse within 6 months
- Claim settlements – lump sum paid in lieu of future monthly payments

List 5 **LTCl-specific** considerations for experience studies.

Source: Experience Study Calculations

1. Elimination period – typically 30, 60, 90, 180 and 365 days and can also vary by LTC claim type
2. Some LTCI benefits indexed to inflation
3. Mortality rates do not follow any established table
4. Claim data must capture **diagnosis** (very important since it drives claim length)
5. Claim data not always organized ideally and requires work

Experience study considerations for **deferred annuities**?

Source: Experience Study Calculations

- Deferred annuity utilization rates
 - GMxBs: compare benefit paid to amount payable in absence of GMxB
- Contract year data challenges
 - Data usually monthly
 - Common approach: convert account balances to contract year basis using 13-month average

Experience study considerations for **payout annuities**?

Source: Experience Study Calculations

- Immediate annuities
 - ▶ Self-selection – people buy IAs when they think they're healthier than average
 - ▶ Choice of IA may indicate health
 - ▶ Larger IAs tend to have lower mortality
 - ▶ Amount exposure should be based on monthly payment, not reserves
 - ▶ Mortality improvement is key consideration
- Structured settlements – arise from injury lawsuits
 - ▶ Usually involves one or more lump sums (not lifetime payouts)
 - ▶ Mortality is higher than IAs
 - ▶ Amount exposure should be based on reserves or premium paid

List the 6 major steps in table development.

Source: Table Development

1. Develop data
2. Identify table dimensions through exploration techniques
3. Populate table through graduation/interpolation or modeling
4. Extend and project rates as needed
5. Review and adjust rates after assessing financial impact
6. Finalize table

List the 4 preliminary activities of data development and analysis.

Source: Table Development

1. **Review previous studies (if they exist)**
2. **Clarify purpose of table and goals**
3. **Ensure confidentiality of each contributor's data**
4. **Review available data** (homogeneity, credibility, etc.)

What is the data call process?

A **data call** is a formal request for data from **contributors**

- Simpler data calls are more successful
- Document what is needed for each **data item** (“columns”)
- Define data **structure** (e.g. relational database)
- Create **detail records** (“rows”) combinations of life, age, etc.
- Document all table relationships within relational databases

List 6 common data challenges.

Source: Table Development

1. Incomplete data
2. Terminology may vary by contributor
3. Data may not arrive on time
4. Wrong or improperly formatted data
5. Time lags in reporting of events like death
6. Lack of resources to transform data into requested layout

List the 6 components in an experience study calculation.

1. **Study anniversary date** – birthday, policy anniversary, etc.
2. **Age basis** (ALB, ANB, etc.)
3. **Experience study summary records**
4. **Count-based and amount-based rates** (q_x 's, etc.)
5. **Expected results** (A/E ratios, etc.)
6. **Summary results for each detail record** (event counts and amounts)

List 3 ways to address distortion from having a mix of age bases in experience data.

Source: Table Development

1. Assume all ages based on most common method
2. Recalculate on common basis for each record
3. Use weighted-average age

List the 8 steps in the data analysis process.

Source: Table Development

1. **Acquire** data
2. Data **validation**, preliminary **exploration**, and **outlier** analysis
3. Data **visualization** and preparation
4. Determine **analytical approach**: exploratory vs. advanced analytics
5. **Model creation** and assessment for advanced analytics
 - ▶ Select model(s) with explanatory ability, predictive power, and implementation ease
6. **Select final model(s)**
7. Minimize table **dimensions**
8. Replace grids with **factors** if possible

List the 3 steps in the modeling process.

Source: Table Development

The modeling process:

1. Model fitting: select variables with lowest p-value
2. Avoid confounding
3. Create transformed variables, functions, interaction terms, or stratification

What is the purpose of graduation?

Graduation is a mathematical process that smooths an array of rates

- Assumes that the “true” rates follow a continuous curve
- Required when rates do not come from models
- Can also be used on modeled rates
- Goal: balance fit and smoothness

List the properties of a preferred graduation method.

Source: Table Development

1. Produces same total events as ungraduated rates
2. Parameters control amount of smoothness applied
 - ▶ Input fit, table fit, smoothness

Credibility and data grouping considerations when graduating rates?

Non-credible (“incredible”) rates distort graduated rates

- Smoothness calculation weights credible and non-credible rates equally
 - Biggest problem: very young and very old ages
- ↑ credibility usually comes at cost of graduation
 - Grouping rates by age (e.g. quinquennial)
 - Grouping data by calendar or policy years
- Possible solution: use exposure-weighted averages for graduation

List the 4 steps of the graduation process.

1. **Collect and populate graduation input**
2. **Review and adjust input as needed**
3. **Run graduation process** for 3 parameters: input fit, table fit, smoothness
 - ▶ Set parameter value and run graduation algorithm
 - ▶ Review/graph/evaluate results
 - ▶ Repeat until satisfied with parameter
4. **Run final graduation** using best fit and smoothness parameters

Formula for total variance for all lives (assuming individual data available)?

Total variance for all lives:

$$\text{Variance}_r = q_r(1 - q_r) \sum \text{Amount}_x^2 \times \text{ExposedCount}_x$$

Ways to approximate total variance for all lives?

Source: Table Development

Approximations for variance (from least refined to most):

1. Use overall average size:

$$\text{Variance}_r \approx \text{AvgSize}_{OA}^2 \times q_r(1 - q_r) \sum \text{ExposedCount}_x$$

2. Use average size for lives contributing to q_r :

$$\text{Variance}_r \approx \text{AvgSize}_r^2 \times q_r(1 - q_r) \sum \text{ExposedCount}_x$$

3. Allocate average size between the 2 most common sizes in size groups
 - Can also be used to develop factors to apply to method 1 or 2

List common interpolation methods for table development.

- **1-dimensional methods**

- ▶ Linear (simplest method)
- ▶ Cubic spline
- ▶ Log-Linear or Log-Cubic – useful for mortality rates ($\uparrow$ at constant rate)

- **2-dimensional methods**

- ▶ Bilinear
- ▶ Bicubic (smoothest)

List 3 methods for extending rates in a table.

Source: Table Development

Often necessary for very young or old ages where credibility is low

1. **Using rates from existing credible table**

- ▶ Use slope as guide
- ▶ Grade from study's rates to existing table rates

2. **Other data sources**

- ▶ SSA for old ages

3. **Formulas**

- ▶ Should reproduce rates for nearby credible ages

List 5 reasons why initial rates developed for a table may be deficient.

Source: Table Development

1. Nonsensical rates (e.g. mortality rate ≥ 1)
2. Data had poor credibility
3. Nonsensical patterns of rates
4. Suspicious differences in arrays of rates
5. Very wide confidence intervals

List the 5 components of enforcement reviews.

Source: Table Development

1. Define relationships to be enforced (agree on when project starts)
2. Define when relationships will be checked
3. Create spreadsheets to check relationships (automation)
4. Check S&U mortality rates
5. Adjust rates to enforce relationships

List 4 considerations for projecting future rates.

Source: Table Development

- Use historical data based on consistent population over time
- Mortality trends are difficult to estimate
- Connect cause and effect
- 3 types of projections
 1. Projecting from mid-point of experience data to final table effective date
 2. Projecting beyond effective date at single rate of improvement
 3. Projecting beyond effective date at varying rate of improvement

Items to consider when assessing financial impact of a table?

- **Reserves and nonforfeiture values**
 - Prescribed industry tables affect prescribed stat reserves and nonforfeiture values
 - PBR can be impacted by internally developed tables
 - Impact can vary significantly from company to company
- **Premiums and policyholder dividends**
 - Could be impacted by new industry or internally-developed tables

List 5 ways to create additional tables from final table rates.

Source: Table Development

Additional tables can be created for other combinations:

1. With or without projected trend factors
2. With or without valuation loading factors
3. Multiple age definitions (ANB, ALB, etc.)
4. Unisex
5. Relative risk versions for UW purposes

List the 4 major steps in the valuation loading process for the commissions' valuation table (CVT).

Source: Table Development

1. Develop **experience table** based solely on experience data
2. Develop the **valuation basic table (VBT)**
3. Apply loading factors to experience table → **loaded experience table**
4. Create the **Commissioners' Valuation Table** (will be used for stat reserves)

Define credibility and its relationship to data volume.

Credibility in actuarial context = **statistical credibility**

- Arose more than a century ago in P&C industry
- **Low credibility** does not mean “not believable”
 - It means “should have less weight”
- Credibility $\uparrow$ as data volume $\uparrow$ ($\sigma \downarrow$)
- **Full credibility** occurs when addition of more data would be immaterial
 - Any remaining error is inconsequential

List the 3 major families of credibility methods.

1. **Limited Fluctuation Method** (aka Classical Credibility)

- ▶ Intended to limit effect of random experience fluctuations on estimate
- ▶ Simple and widely used, but has theoretical shortcomings

2. **Greatest Accuracy Method** (aka Bühlmann Credibility)

- ▶ Appeared in the 1960s
- ▶ Intended to make estimation errors as small as possible
- ▶ More complex and less popular, but more theoretically grounded

3. **Bayesian Method**

- ▶ Combines prior distribution with likelihood function based on observed data
- ▶ Also the foundation of GAM
- ▶ Has not been used much, but usage may ↑ in the future

General formula for determining credibility-weighted rate?

$$\text{Credibility-Weighted Rate} = \begin{cases} Z \times \text{Observed Rate} + (1 - Z) \times \text{Prior Rate} & \text{(LFM)} \\ Z \times \text{Observed Rate} + (1 - Z) \times \text{Portfolio Rate} & \text{(GAM)} \end{cases}$$

Observed Rate = best estimate or result from recent experience study

Z = weight assigned to **Observed Rate**

- Ranges from 100% (fully credible) to 0% (no credibility)

$1 - Z$ = remaining weight assigned to:

- **Prior Rate** (LFM) – prior rate based on much larger experience study
 - Assumed to be fully credible
 - Often based on large industry table (e.g. 2001 VBT)
- **Portfolio Rate** (GAM) – overall rate for a set of related risk groups

Prior rate used for Limited Fluctuation Credibility?

Prior rate = most appropriate estimate absent any experience data

- Should be consistent with the business in scope
- Potential problems:
 - ▶ May be based on different target market, UW, and/or distribution system
 - ▶ Requires adjustments or else will introduce error
 - ▶ May not be fully credible

Credibility weight formula under Limited Fluctuation Credibility?

$$Z = \min \left[1, \sqrt{\frac{N_{AC}}{N_{FC}}} \right]$$

- N_{AC} = number of actual observations (claims, policies, etc.)
- N_{FC} = number of observations required for full credibility
- P_{FC} = probability threshold for full credibility
- R_{FC} = error tolerance for full credibility

Most common assumption: $P_{FC} = 90\%$, $R_{FC} = 5\%$

- We want our rate to be within $\pm 5\%$ of true rate at least 90% of the time
- $N_{FC} = 1082$ for normally distributed claims (P&C, health)
- $N_{FC} = 1082p = 1082(1 - q)$ for binomially distributed claims (mortality, lapse)

Relationship between observed rate and full credibility?

Source: Credibility Methods Applied

- As $q \uparrow$, $N_{FC} \downarrow$
- Credibility is based on relative error: as $q \uparrow$, error threshold $\uparrow$
- Small decrement studies require more exposed lives (n)

Asymptotic formula for credibility?

Developed by Albert W. Whitney (1918):

$$Z = \frac{N}{N + K}$$

where N = exposure (e.g. claim count) and K = “a constant”

- $Z = 0$ for no experience data and $Z \rightarrow 1$ as $N \rightarrow \infty$
- K can be determined by actuarial judgment or based on experience
- Higher values of K result in:
 - ▶ Lower credibility weights
 - ▶ Slower convergence as $N \rightarrow \infty$

How to calculate credibility weights based on confidence intervals?

Source: Credibility Methods Applied

Z = probability that observed rate is within “tight confidence interval”

$$Z = \Pr [\mu^T(1 - r) \leq \mu \leq \mu^T(1 + r)]$$

- μ^T, σ^T = true (unknown) values of mean and standard deviation
- μ, σ = observed values
- r = ratio that specifies the tight C.I. (analogous to R_{FC})

Assuming standard normal distribution and $\mu/\sigma \approx \mu^T/\sigma^T$:

$$Z = \Pr \left[-r \frac{\mu}{\sigma} \leq \frac{(\mu - \mu^T)}{\sigma^T} \leq r \frac{\mu}{\sigma} \right]$$

Credibility increases when $\mu/\sigma \uparrow$ and/or $r \uparrow$ (wider C.I.)

Normal distribution appropriate for large N ; binomial is best for mortality/lapse

How to calculate number of deaths required for full credibility in mortality study?

Source: Credibility Methods Applied

1. Calculate observed mortality rates by band and in aggregate

$$q_{\text{cnt}} = \frac{\text{Count of Deaths}}{\text{Count Exposed}} \quad q_{\text{amt}} = \frac{\text{Amount of Deaths}}{\text{Amount Exposed}}$$

2. Calculate variance (σ^2) and standard deviation (σ)

$$\sigma_{\text{cnt}}^2 = (\text{Count Exposed})(q_{\text{cnt}})(1 - q_{\text{cnt}})$$

$$\sigma_{\text{amt}}^2 = (\text{Count Exposed})(\text{Policy Size})^2(q_{\text{amt}})(1 - q_{\text{amt}})$$

3. Calculate standard deviation as % of deaths observed

$$\text{Std Dev as \% of Deaths} = \frac{\sigma}{\text{Count of Deaths}}$$

$$\text{Full Cred Std Dev as \% of Deaths} = \frac{0.05}{1.64485} = 3.04\%$$

4. Calculate deaths required for full credibility (count and amount)

$$\text{Study to Full Cred Ratio} = \frac{\text{Std Dev as \% of Deaths}}{\text{Full Cred Std Dev as \% of Deaths}}$$

$$\begin{aligned}\text{Full Cred Deaths} &= (\text{Study to Full Cred Ratio})^2 \times \text{Observed Deaths} \\ &= 1082p \text{ for count-based}\end{aligned}$$

How does credibility differ on count vs. amount basis?

- **Effect of policy count distribution by face amount size:**
 - ▶ Smaller policies usually outnumber larger policies
 - ▶ Therefore count-based exposure is higher at lower face amounts
 - ▶ Count-based $\sigma^2 \downarrow$ as policy size $\uparrow$ (opposite for amount-based)
 - ▶ Count-based variance is directly proportional to sample size: npq
 - ▶ Amount-based variance = $npq \times \text{Size}^2$
 - ▶ Takeaway: count-based Z may overstate credibility on larger policies
- **Amount-based credibility weights tend to be lower than count-based**
 - ▶ Main reason: credibility $\downarrow$ as variance $\uparrow$
 - ▶ Amount-based variance is usually higher than count-based

Goal and assumptions underlying Greatest Accuracy Credibility?

Goal: minimize expected value of square of differences between:

1. Bühlmann-weighted estimate
2. Bayesian predictive mean

GAM makes 2 assumptions:

1. Each risk group in portfolio follows its own unknown distribution
2. The distributions of the groups follow an unknown distribution

Greatest Accuracy credibility weight formula?

Source: Credibility Methods Applied

$$Z = \frac{N}{N + K} = \frac{N}{N + v/a}$$

N = claim count exposure

Portfolio = collection of related risk groups

- Groups of similar risks from different geographical areas or occupations
- Collection of insurance companies

v = expected value of process variance = $E[\text{Var}[X]]$

- As $v \downarrow$, credibility $\uparrow$
- $v \downarrow$ when risk group has large exposure or claim counts

a = variance of hypothetical means = $\text{Var}[E[X]]$

- As $a \uparrow$, credibility $\uparrow$
- $a \uparrow$ when there is more variation in means
- Extreme case: if $a = 0$, $Z = 0$ (nothing unique about each group)

List strengths and weaknesses of Limited Fluctuation Credibility.

Strengths of LFM:

- Simple and easy: only requires number of claims
- Useful for experience rating when prior rate is appropriate
- Only requires data for risks being assessed and prior rate
- Could be improved by using σ as % of claims to calculate Z under C.I. method

Weaknesses of LFM:

- Formulas and inputs are arbitrary
- Prior rate gets same weight, $1 - Z$, regardless of its accuracy
- May significantly overstate count-based Z 's if rates are based on amount
- Results distorted if there is no prior rate (e.g. new product)
- Variance may be worse measure of “quality” than MSE used by GAM
- Z reaches 1.0 prematurely compared with asymptotic formula

List strengths and weaknesses of Greatest Accuracy Credibility.

Strengths of GAM

- Spells out all aspects and clearly states all assumptions
- Produces exact answer by applying basic probability principles
- Has clear objective function: minimize mean-square error
- No arbitrary choices unrelated to observed random variables

Weaknesses of GAM

- Requires portfolio of data from comparable risk groups
- Portfolio rate will be distorted if risk groups not sufficiently similar
 - Analogous to prior rate not being appropriate under LFM
- May be inaccurate when random variable has heavy tail
- Empirical Bayesian estimates of σ^2 can be negative, “which is nonsensical”

Compare Greatest Accuracy Credibility with Bayesian Credibility.

Bayesian credibility was starting point for Bühlmann credibility (GAM)

- GAM credibility-weighted rate falls between observed rate and portfolio rate
- Bayesian: not necessarily

GAM has the following advantages over Bayesian:

- Requires less mathematical skill
- Computationally easier
- Does not require prior distribution
 - GAM uses experience data instead

List at least 4 characteristics of lapse-supported insurance.

Source: ILA101-107: Lapse Supported Insurance

1. PV profits assuming zero lapses ; PV profits with planned assumption
 - ▶ ASOP 24: can't show illustrations if policy fails self support test
2. Common features: guaranteed level premiums, low/zero CSV, high late benefits
3. Premiums too low → insufficient reserves
 - ▶ Warning sign: higher-than-expected early profits
4. Risk is in ultimate lapses, not early lapses
 - ▶ Lower-than-expected early lapses ↑ profit
 - ▶ Lower-than-expected ultimate lapses ↑ benefit costs
5. High income market is highest risk
 - ▶ Competitive/marketing pressures keep premiums low
 - ▶ Policyholders least likely to lapse
 - ▶ Agent persistency data may be illusion due to past replacement activity

How return-of-premium riders affect lapse risk?

- ROP riders create tontine – everyone pays in, but only “survivors” collect ROP
 - Bad investment for policyholder on lapse-adjusted basis
- ROP cash value not enough to lower risk of low lapses
- ROP can convert limited-pay products to lapse-supported
- Benefits of ROP:
 - Less risky if lapses driven by independent events (e.g. employee turnover)
 - Help blocks/companies with poor persistency
 - ↑ profitability in low income markets

List key findings from SOA study on flexible premium UL premium persistency.

- **Many companies simply assume 100% premium persistency**
 - Aggregate premium falls with duration due to limited pay business
 - Companies often vary persistency by duration
- Pricing assumptions vs. CFT and GAAP/IFRS
 - **Pricing assumes more patterns in general**
 - Some companies use weighted average persistency assumption
 - Companies adjust ULSG pricing persistency to prevent lapses
- **Areas for improvement** (few companies currently doing):
 - Dynamic premium persistency assumptions
 - Sensitivity testing

List types of post-level premium structures in North America.

Source: Selective Lapsation for Renewable Term

Post-level premium structures in the US:

1. Jump to ART – premium jumps from level to much higher ART premiums
2. Grade to ART – premium gradually ↑ from level to ART
3. Jump to new level period
4. Jump to other – combination of 2 and 3

In Canada, most products are “jump to new level”

Selective Lapsation Provisions from CIA Education Note “Expected Mortality”

Source: Selective Lapsation for Renewable Term

- **Selective lapses** – have same mortality as newly selected lives ($q_{[x+t]}$)
- Lapse rates $\uparrow$ temporarily when premium rates $\uparrow$ at renewal date
- Healthy lives more likely to lapse at renewal than unhealthy lives
- Factors that drive level of selective lapsation:
 - ▶ Size of premium rate increase
 - ▶ Period between premium increases
 - ▶ Duration
 - ▶ Policy size
 - ▶ Distribution systems
- Selective lapses may occur at times other than at renewal
- PL mortality experience may not reflect selective lapses (adjust as needed)

SOA Survey on Post-Level Assumptions

Source: Selective Lapsation for Renewable Term

- Most insurers assumed $\downarrow$ 100% lapse rate at end of level period
- Assumed shock lapse $\uparrow$ as level period $\uparrow$
 - 100% more common for 20-30-yr term than 10- and 15-yr
 - Median shock lapse: $\sim$ 80% for 10-yr to $\sim$ 95% for 30-yr
- Median PL mortality = 230%–300% of level period
- Mortality deterioration assumptions generally grade down over 3 years
- Assumed lapses and mortality deterioration had strong correlation
- Median premium jump = 3x–25x
 - Varied by issue age, gender, term period, and risk class

SOA Report on Post-Level Experience: Lapse

Source: Selective Lapsation for Renewable Term

Lapse dates set equal to beginning of grace period if they occurred during grace period

- Grace period = time after missed insurance premium is due

Key lapse findings:

- Shock lapses ↑ with premium jump ratio (PJR)
- Lapse trends by PJR are similar across companies:
 - ▶ ↑ very quickly at lowest PJRs
 - ▶ ↑ at decreasing rate with PJR
 - ▶ Level off at highest PJRs
- Within each PJR band, lapses ↑ with issue age and face amount
- Lapses heavily skewed to end of final level year and beginning of first PL year

SOA Report on Post-Level Experience: Mortality

Source: Selective Lapsation for Renewable Term

- Mortality rates $\uparrow$ significantly with PJR
- Higher lapses $\uparrow$ effect of excess mortality in grace period
 - Provides “free” life insurance to policies that have lapsed
- Mortality $\uparrow$ at increasing rate as lapse rates $\uparrow$

Describe the subpopulations within a cohort of lives at beginning of the post-level period.

Source: Selective Lapsation for Renewable Term

$$T = U + A + S + P = 100\% = \text{total cohort}$$

- S = excess lapsers who have select mortality (healthy lives)
 - Often unknown (requires estimation and sensitivity testing)
 - $S = \text{Selective\%} \times (\text{Total Lapse Rate} - \text{Base Lapse Rate})$
- A = excess lapsers with average mortality
 - Extra lapses that occur for reasons other than health
 - $A = (1 - \text{Selective\%}) \times (\text{Total Lapse Rate} - \text{Base Lapse Rate})$
- U = lapsers with same mortality as underlying (base) mortality
 - Normal, base lapses (not lapsing for health reasons)
- P = persisting cohort (policyholders that don't lapse)
 - Generally unhealthy lives willing to pay high premiums to keep policy inforce
 - We can solve for $P = 1 - S - A - U$
 - Since only P remains inforce, total lapse rate = $S + A + U$

Describe 3 primary methods for estimating post-level mortality.

Source: Selective Lapsation for Renewable Term

1. **Dukes-MacDonald**

- ▶ Originally assumed all excess lapses are select
- ▶ Later variations (DM1, DM2) assume some excess lapses are non-select

2. **Becker-Kitsos**

- ▶ Refines DM by adding effectiveness factor
- ▶ Assumes additional lapses are fully select plus extra amount of mortality
- ▶ Extra mortality grades off over select period

3. **VTP2:** published by CIA in 1986 Valuation Technique Paper

- ▶ Similar to DM1, but with following differences:
 - ▶ A and S lapse rates apply just prior to anniversary
 - ▶ Underlying (U) lapse and mortality rate apply continuously
- ▶ DM1 assumes U lapses occur immediately prior to S lapses

Formula for persisting mortality under Becker-Kitson, Dukes-MacDonald (DM1 and DM2), and VTP2?

Source: Selective Lapsation for Renewable Term

$$q'_{[x]+t} = \begin{cases} S \times (q_{[x+t]} + q_{[x+t]}^{\text{extra}}) + U \times q'_{[x]+t} + (1 - S - U) \times q''_{[x]+t} & \text{for BK} \\ S \times q_{[x+t]} + A \times q_{[x]+t} + U \times q'_{[x]+t} + (1 - S - A - U) \times q''_{[x]+t} & \text{for DM1} \\ S \times q_{[x+t]} + U \times q'_{[x]+t} + (1 - S - U) \times q''_{[x]+t} & \text{for DM2} \\ S \times q_{[x+t]} + A \times q_{[x]+t} + (1 - S - A) \times q''_{[x]+t} & \text{for VTP2} \end{cases}$$

Under BK:

- $q_{[x+t]}^{\text{extra}} = R \times G(t) \times (q_{[x]+t} - q_{[x+t]})$
- Recommend assuming $G = 1$

Describe common problems/issues for all persisting mortality methods.

Source: Selective Lapsation for Renewable Term

- **Skewness: all methods assume excess lapses occur EOY**
 - Reality: ~90% of lapses occur BOY right after renewal
 - Underestimates mortality deterioration
- **Shape of underlying mortality table**
 - If industry table multiples too low, excess mortality > 0
 - Excess mortality = $q'' - q'$
 - Ideally, excess mortality should grade to zero over time
- **Accounting for grace period deaths**
 - More important when lapse rates are high
 - DM1 mortality for persisters who die in grace period:

$$\left[S \times q_{[x+t]} + (A + U) \times q'_{[x]+t} \right] \times \frac{\text{Grace Period}}{365 \times (1 - T)}$$

- $T = S + A + U$ = total lapse rate occurring at renewal date
- Add this extra mortality to PL persister mortality
- Under VTP2, formula is same with $U = 0\%$

How appropriate are the main persisting mortality methods for US and Canadian experience?

Source: Selective Lapsation for Renewable Term

Key conclusions were similar based on US and Canadian experience:

- VTP2 doesn't replicate actual post-renewal mortality unless you use very high S
- DM1 comes closer to matching actual mortality experience
- Caveats:
 - ▶ Credibility is low, so use caution with results
 - ▶ No Canadian data reflects both mortality and lapse

How can VTP2 be revised to be more accurate?

Source: Selective Lapsation for Renewable Term

Original VTP2 calculated selective and average lapse rates incorrectly

$$S = (w^{\text{total}} - U) \times \text{Selective\%}$$

$$A = (w^{\text{total}} - U) \times (1 - \text{Selective\%})$$

The correct calculation reflects that excess lapse rates occur after underlying lapses:

$$S = \frac{(w^{\text{total}} - U) \times \text{Selective\%}}{1 - U}$$

$$A = \frac{(w^{\text{total}} - U) \times (1 - \text{Selective\%})}{1 - U}$$

“VTP2-revised” results in higher select and average lapse rates

- And therefore **higher persister mortality** since $q'' \uparrow$ as $P \downarrow$
- q'' was as much as 22% higher after correcting lapse rates

List 3 philosophies for covering cost of conversions.

Source: Term Conversions: Pricing and Reserves

1. **Reflect conversion cost in term pricing** (most common)
 - ▶ *This is what the author advocates*
 - ▶ Allows reserves to build for conversions
 - ▶ Aligns revenue with source of risk (term “swaption”)
2. **Develop separate permanent premiums for conversions**
 - ▶ Avoids cross-subsidization but is administratively challenging
 - ▶ Alternatively, could reinsure conversions using YRT PISM rates
3. **Reflect extra conversion mortality in permanent policy pricing**
 - ▶ Might be appealing if insurer has mature experience
 - ▶ Can make permanent mortality artificially high

Two stage model for pricing term conversions?

Source: Term Conversions: Pricing and Reserves

1. At issue, determine expected extra mortality cost for term conversions

- For each year when conversion is possible calculate:

$$\begin{aligned}\text{Extra NSP for Conversions} &= K_{x,r} = \text{PVFB}^{\text{Converted}} - \text{PVFB}^{\text{Standard WL}} \\ &= \text{reserve for conversions at duration } r\end{aligned}$$

2. Calculate charge for term policyholders

$$\begin{aligned}A_{x,r} &= {}_r p_x \times e_{x,r} \times K_{x,r} \times v^r \\ \text{Total PV of Expected Excess Mortality} &= \sum_r A_{x,r} \\ \text{Level Conversion Charge} &= \frac{\text{Total PVFB of Excess Mortality}}{\ddot{a}_{x:\overline{n}|}}\end{aligned}$$

Factors that cause cost of conversions to vary?

Conversion cost ↑ when base mortality ↑

- Higher issue ages
- Males (compared to females)
- Smokers (compared to non-smokers)
- Non-preferred (compared to preferred classes)

Relationship between conversion period length and conversion cost?

Restricting conversion period reduces cost of conversion

- Disallowing conversions after year 10, reduces cost roughly 30%
- Caveat: policyholders will accelerate conversions in response

Compare newer statistical approaches with traditional experience studies.

Advantages of statistical approaches over traditional experience studies:

- Consider all variables simultaneously (eliminates bias in correlated variables)
- Handles data more efficiently
- Generates smoother curves without manual adjustments
- Provides distribution of outcomes rather than only mean
- Cross-term variables can capture interrelationships and improve model fit

Key disadvantage of statistical models: can become complex

- Simpler models easier to interpret and implement in practice
- Actuary must balance complexity/accuracy with simplicity

List key factors that cause conversion rates to vary.

- **Late converters have highest conversion rates**
 - Year 10 conversion rate $\approx$ 7% compared to $\downarrow$ 1% in years 1–9
- **Year 10 conversion rates higher for worse risk classes**
 - Smokers $\downarrow$ Standard NS $\downarrow$ Preferred NS
 - Interpretation: smokers more likely to be declined if applying for new policy
- **Impact of issue age varies by duration:**
 - Early years: youngest issue ages have slightly higher conversion rates
 - Middle years: no difference by issue age
 - Years 9–10: older issue age $\rightarrow$ higher conversion rates
- **Large policies have highest year 10 conversion rates**
 - A 1M policy's conversion rate is nearly 2x a $\downarrow$ 100K policy

List key factors that cause mortality to vary for converted policies.

- **PISM ratios highest 1 year after conversion, then fall**
 - High mortality immediately after conversion indicates anti-selection
 - Even 10 years later, PC mortality > 100% of level period
- **Larger policies have higher PC mortality** (more anti-selection)
- **Late converters have highest PISM ratios after conversion**
 - Late converter PISM ratios decline after year 1
 - Middle converter PISM ratios slightly higher than early converters
 - Middle and early converter PISM ratios are flat after conversion
- **Younger issue ages have higher PISM ratios after conversion**
 - <40 group's initial PISM ratio >500%
 - Next highest group: 60+ (300%)
 - All PISM ratios decline by year after conversion

List key factors that cause lapse rates to vary for converted policies.

- **Overall lapse rates highest 1–2 years after conversion**
 - PC lapse rates roughly 5% and 5.5% 1 and 2 years after conversion
 - Lapse rates fall to about 2.5% by year 4, then level off
- **Lapse rates by gender fairly similar all years after conversion**
 - Female lapses slightly higher initially; reverses later
 - Gender may not be useful model variable
- **Late converters have lower lapse rates after conversion**
- **Larger policies have lower PC lapse rates**

List most significant drivers of anti-selection on term conversions.

Anti-selection highest for late converters

- Highest conversion rates
- Highest initial post-conversion mortality ratios
- Lowest post-conversion lapse rates

Other factors that ↑ anti-selection risk:

- Younger issue ages
- Larger policies
- Smokers